

上海交通大学试卷 (A 卷)

(2020 至 2021 学年 第 1 学期)

班级号 _____ 学号 _____ 姓名 _____
课程名称 _____ CS124 计算导论 _____ 成绩 _____

Python 3 is the only programming language allowed for this course. **100 points in all.**

(1). Please choose the correct answer: Only one is correct for each question. (2 points each, 40 points.)

1. What is the output of the following code? ()
`print([] is [], []==[])`

A. True True B. True False C. False True D. False False

2. Which code does not have error(s) when running? ()

A.
`S = "{0} {0}".format([1])`

B.
`S = "123"`
`S[0] = "0"`

C.
`from math import tan`
`print(math.tan(2))`

D.
`fl = open("1.txt", "w")`
`fl.write(123)`
`fl.close()`

3. Given the code `print(f(lamba x,y:x+y,[1,2,3,4,5]))`, the output is 15, what is 'f'? ()

A. map B. add C. reduce D. filter

4. What is the output of the following code? ()
`print({x:y for x in range(2) for y in range(2)})`

A. {0: 1, 1: 1} B. {0:1, 1:0} C. {0:0, 0:1, 1:0, 1:1} D. {1:0, 1:1}

5. What is the output of the following code? ()

```
A = [[0,1,2,3]] * 3
A[0][0] = 4
del A[-1]
print(A, end="")
```

A. [] B. [[4, 1, 2, 3], [0, 1, 2, 3]] C. [[4, 1, 2, 3], [4, 1, 2, 3]] D. [[0, 1, 2, 3], [0, 1, 2, 3]]

我承诺，我将严格遵守考试纪律。

承诺人：_____

题号	1	2	3	4	5					
得分										
批阅人(流水阅卷教师签名处)										

6. What is the returned value of the function showed below when the function parameter is a python list [1, 2, 3] ? ()

```
def f(s):
    if s == []:
        return []
    return f(s[1:]) + [x + [s[0]] for x in f(s[1:])]
```

f([1, 2, 3]) = ?

- A. [], [1], [2], [1, 2], [3], [1, 3], [2, 3], [1, 2, 3]]
- B. [], [3], [2], [3, 2], [1], [3, 1], [2, 1], [3, 2, 1]]
- C. [[3, 2, 1], [3, 2], [3, 1], [3], [2, 1], [2], [1], []]
- D. [[1, 2, 3], [2, 3], [1, 3], [3], [1, 2], [2], [1], []]

7. Given the definition of function **f** and the lambda expressions **add**, **minus**, **multi** and **power**, what is the returned value of calc([1,2,3,4,5], add, power, minus, multi) ? ()

```
add = lambda x, y: x + y
minus = lambda x, y: x - y
multi = lambda x, y: x * y
power = lambda x, y: x ** y
```

```
def calc(operands, *operations):
    for op in operations:
        operands.append(op(operands.pop(), operands.pop()))
    return operands[-1]
```

- A. 115 B. 162 C. 290 D. 727

8. What is the output of the python code below? ()

```
i = 0
L = [1, 2, 3]
i, L[i] = L[i], i
print(i, L, end="")
```

- A. 1 [0, 2, 3] B. 1 [1, 0, 3] C. 1 [1, 2, 3] D. 1 [1, 1, 3]

9. The output of the following program is:

()

```
def f():  
    s = 100  
    s += 200  
    print(s)  
    del s  
    print(s)
```

s = 0

f()

A. 0

B. 100

C. 200

D. Error

10. The following part is the definition of

class T:

def __init__(self, a, b):

self.a = a

self.b = b

What will the following codes output after their execution?

()

a = T(1, 1)

b = a

b.a, a.b = 2, 3

print(a.a, b.b)

A. 1 1

B. 1 3

C. 2 1

D. 2 3

11. What is the output of the python code below?

()

alist = [1, 2, "def", 3, "abc"]

b = (x for x in alist)

next(b)

aset, atuple = set(b), tuple(b)

print(aset, atuple)

A. {2, 3, 'def', 'abc'}, ()

B. {1, 2, 3, 'def', 'abc'}, ()

C. {2, 3, 'def', 'abc'}, (1, 2, "def", 3, "abc")

D. {1, 2, 3, 'def', 'abc'}, (1, 2, "def", 3, "abc")

12. Given the following codes, which output is correct?

()

x = ['ab', 'cd']

for i in x:

 x.append(i.upper())

print(x)

A. ['ab', 'cd']

B. ['AB', 'CD']

C. ['ab', 'cd', 'AB', 'CD']

D. No output

13. What is the result of 24 / 2 ** 4 in Python3?

()

A. 2

B. 1

C. 1.5

D. 20736

14. Which of the following is valid? ()

A.

```
a = ("a", "b", "c")
a[0] = "2020"
```

B.

```
b = ["a", "b", "c"]
b[-4] = "2018"
```

C.

```
c = ["a", "b", "c"]
c[-4:] = "d"
```

D.

```
d = "abc"
d[0] = "d"
```

15. s = "the sky is blue", the result of the expression print(s[-4:], s[:-4]) is: ()

A. blue the sky is

B. blue is sky the

C. sky is blue the

D. the sky is blue

16. Choose the correct output for the following program: ()

```
sentence = "hello 54749110 this is john"
counter1, counter2, counter3 = 0, 0, 0
for character in sentence:
    if character.isdigit():
        counter1 += 1
    if character.isalpha():
        counter2 += 1
    else:
        counter3 += 1
print(counter1, counter2, counter3)
```

A. 8, 15, 12 B. 8, 15, 4 C. 8, 15, 5 D. 8, 16, 4

17. Choose the correct output for the following program: ()

```
def mypow():
    plist = []

    for i in range(3):
        plist.append(lambda x: x**i)

    return plist

for i in range(3):
    print(mypow()[i](2), end=" ")
```

A. 4 4 4

B. 1 2 4

C. 2 2 2

D. 1 1 1

18. Choose the correct output for the following program: ()

```
from functools import reduce
alist = (4, 3, 7, -1)
```

`print(reduce(lambda a, b: (a,b)[a<=b], alist))`

A. 4 B. 3 C. 7 D. -1

19. Which value can be the key of dict?

()

① [1,2,3,4] ② (1,2,4) ③ {2:4} ④ "4"

A. ①②③④ B. ①②④ C. ②④ D. ④

20. There is a string S = "12345", which code **CANNOT** generate ["12345"]?

()

① list(S) ② S.split() ③ [S.join("")]

A. ①② B. ①③ C. ② D. ③

Problem 2 (15pts) **Happy Number**

Write an algorithm to determine if a number is "happy".

A happy number is a number defined by the following process: Starting with any positive integer, replace the number by the sum of the squares of its digits, and repeat the process until the number equals 1 (where it will stay), or it loops endlessly in a cycle which does not include 1. Those numbers for which this process ends in 1 are happy numbers.

Example:

Input: 19

Output: True

Explanation:

$$1^2 + 9^2 = 82$$

$$8^2 + 2^2 = 68$$

$$6^2 + 8^2 = 100$$

$$1^2 + 0^2 + 0^2 = 1$$

def is_happy_number(n): # return True or False

Problem 3 (15pts) **Valid Anagram**

An anagram is a word or phrase formed by rearranging the letters of a different word or phrase, typically using all the original letters exactly once. For example, "evil" = "vile".

Given two strings s and t , write a function to determine if t is an anagram of s .

Example 1:

Input: $s = \text{"anagram"}, t = \text{"nagaram"}$

Output: true

Example 2:

Input: $s = \text{"rat"}, t = \text{"car"}$

Output: false

Note: You may assume the string contains only lowercase alphabets.

def is_anagram(s, t): # return True or False

Problem 4 (15pts) **Intersection of Two Arrays II**

Given two arrays, write a function to compute their intersection.

Example 1:

Input: nums1 = [1,2,2,1], nums2 = [2,2]

Output: [2,2]

Example 2:

Input: nums1 = [4,9,5], nums2 = [9,4,9,8,4]

Output: [4,9]

Note:

- ☐ Each element in the result should appear as many times as it shows in both arrays.
- ☐ The result can be in any order.

def intersections(list1, list2): #list1 and list2 are of the type list, return list

Problem 5 (15pts) Hamming distance

The Hamming distance between two integers is the number of positions at which the corresponding bits are different (即对应的二进制表示，有多少位不同)

Given two integers, calculate the Hamming distance.

Input: $x=1$, $y=4$

Output: 2

Because $x=1=001$, $y=4=100$, so the number of different bits of x and y is 2.

def hamming_distance(x, y): # return distance